

Enhancing Khmer Optical Character Recognition By Using Fine-Tuning Tesseract Focus On Specific Font

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Abstract

Khmer Optical Character Recognition (OCR) remains a challenging task due to the script's complexity especially with overlapping vowels and compound characters. While existing OCR engines like Tesseract have made progress in supporting non-Latin scripts and their performance on Khmer documents especially those using standard governmental fonts often yields high error rates. This study focuses on enhancing OCR accuracy for Khmer script by fine-tuning the Tesseract OCR engine on a controlled dataset generated using the Khmer OS Siemreap font. A dataset of 9,315 images of dataset was used to train a Long Short-Term Memory (LSTM) model with adjusted training parameters including increasing the max_iterations from 10,000 to 15,000. The evaluation compares the fine-tuned model against the default Tesseract model using a test set of 13,965 characters containing overlapping vowels and complex combinations. The fine-tuned model achieved a Character Error Rate (CER) of 0.35% its significantly outperforming the default Tesseract model with a CER of 0.88%. These results demonstrate that font-specific fine-tuning can substantially improve Khmer OCR accuracy in real-world document scenarios especially in official texts from ministries and institutions. Future work will explore multi font training and more complex layouts to further generalize OCR performance for Khmer.

1 Introduction

Optical Character Recognition (OCR) is crucial for digitizing printed and handwritten documents by converting images of text into machine-readable text. While OCR systems achieve high accuracy for Latin scripts but Khmer script remains a challenge due to its complex structure and the structure involving overlapping vowels, diacritics and non-linear character arrangements. As highlighted [11] current OCR engines like Tesseract struggle with Khmer's visual complexity. [2] demonstrated significantly better performance than the default Tesseract on Khmer text, confirming the need for script-specific approaches.

Despite efforts like the [8, 14] the existing datasets often lack font and layout diversity found in real-world documents. To address this, the study Fine-Tunes Tesseract OCR using a controlled dataset based on the widely used Khmer OS Siemreap font. The research focuses on printed text with clear backgrounds aiming to develop a model optimized for overlapping characters evaluate performance using Character Error Rate (CER), analyze recognition errors and compare results with both default and 10k-iteration Tesseract models.

This targeted approach creates a controlled baseline for Khmer OCR, isolates structural challenges and contributes useful datasets and insights for future work. The results show that font-specific fine-tuning significantly improves accuracy enhancing Khmer document digitization in low-resource settings. The thesis is organized into five chapters: Introduction, Literature Review, Methodology, Results and Discussion and Conclusion. Chapter 1 outlines the research context, problem and significance. Chapter 2 reviews existing OCR techniques and Khmer-specific studies. Chapter 3 details dataset generation and training with Tesseract. Chapter 4 evaluates and discusses results. Chapter 5 concludes the study.

2 Literature Review

2.1 Background of Related Works

Khmer OCR development has evolved from rule-based and early machine learning approaches to modern deep learning models. While Tesseract especially with LSTM improved printed Khmer text recognition, still struggles with complex scripts without retraining. Recent deep learning models like CNNs, RNNs and Seq2Seq have shown significant performance gains. For example [3] achieved a 0.7% CER using Seq2Seq compared to Tesseract's 35.9%. The availability of datasets like [8] has further advanced Khmer OCR research.

2.2 Traditional OCR Approaches for Khmer Script

Traditional OCR development for Khmer script has faced major challenges due to its complex structure involving numerous consonants, vowels, diacritics and overlapping characters. Early approaches focused on handcrafted features and classical machine learning algorithms. Chey Chanoen applied Legendre Moment Descriptors for recognizing 10 Khmer consonants with 92% accuracy and later enhanced the system using Wavelet Descriptors to support multi-font and multi-size recognition achieving up to 92.85% accuracy across 10 fonts and three font sizes. Ing Leng Ieng employed Hidden Markov Models combined with Discrete Cosine Transform features, reporting accuracies above 98% on Limon fonts. [5] Kruy Vanna's font- and size-independent system used Generic Fourier Descriptors to capture frequency domain features invariant to translation, rotation, and scaling, reaching 97.9% accuracy across 19 fonts.

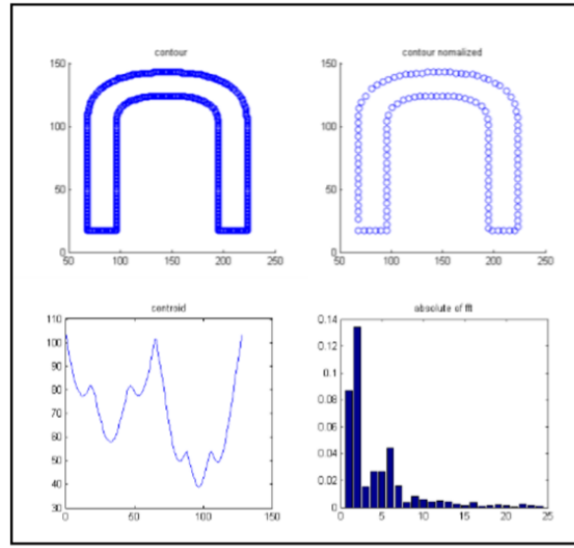


Figure 1: FOURIER DESCRIPTORS Extraction Process [5]

Further improvements came from combining edge detection and template matching techniques by Ieuch SETHA and Taing Nguonly which achieved 99% accuracy on the Khmer OS Content font at 36pt size. Pongsametry Sok and Taing Nguonly introduced Support Vector Machines for printed Khmer recognition, obtaining over 98% accuracy on varying font sizes. Additionally, Boiangiu, Ioanimescu and Dragomir proposed a voting-based OCR system that fused results from multiple algorithms across all OCR stages, significantly enhancing recognition robustness and accuracy.

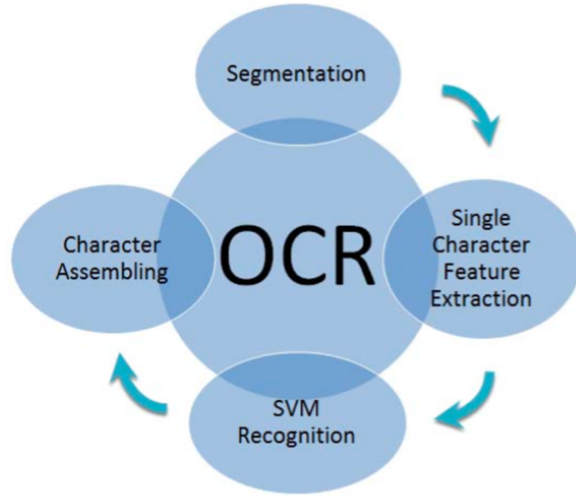


Figure 2: Optical Character Recognition (OCR) Features [12]

2.3 Tesseract OCR and Its Performance on Khmer

Tesseract is a widely used open-source OCR engine supporting over 100 languages including Khmer language. While it performs well on Latin scripts but its effectiveness on Khmer is limited with overlapping diacritics and complex structures. Studies show high Character Error Rates (CER) when using pre-trained Tesseract models on Khmer text, especially with uncommon fonts or layouts.

According to [13] achieving 90% accuracy for Tesseract’s architecture uses a convolutional layer followed by recurrent layers and Connectionist Temporal Classification (CTC) loss to enabling text-line recognition (Liebl & Burghardt, 2020). Mahoney et al. [6] demonstrated the effectiveness of CNN and LSTM models for complex layouts to support Tesseract’s LSTM-based sequence modeling, while Buoy et al. [2] developed a Seq2Seq attention model for Khmer OCR trained in seven fonts that achieved a 1% CER on 3,000 images and outperformed Tesseract’s 3% CER.

According to [4] compared Tesseract to fine-tuned models and found Tesseract’s CER significantly higher, especially due to stacked modifiers, overlapping vowels, and lack of word boundaries. Their “Convex Hull of Languages” visual placed Khmer among similarly complex non-Latin scripts, emphasizing the need for tailored models.

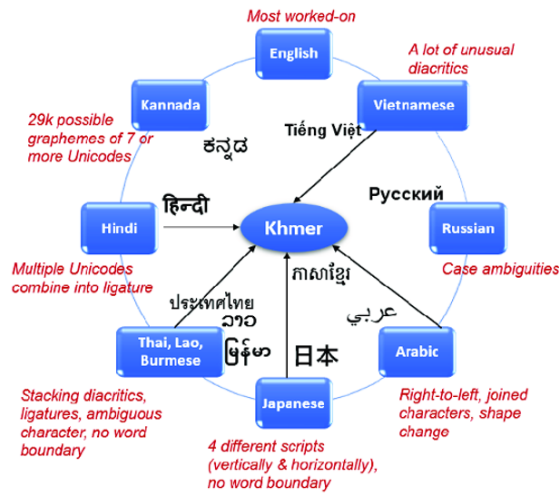


Figure 3: Relations Between the Khmer Script and Other Non-Latin Scripts [4]

EKYC Solutions created a standardized benchmark dataset for Khmer OCR including both clean and complex scanned images. It allows for consistent evaluation of Khmer OCR models including

Tesseract. Smith (2007) in the original introduction of Tesseract emphasized its modular and extensible architecture. Later updates—including LSTM integration—enabled its use on complex scripts like Khmer.

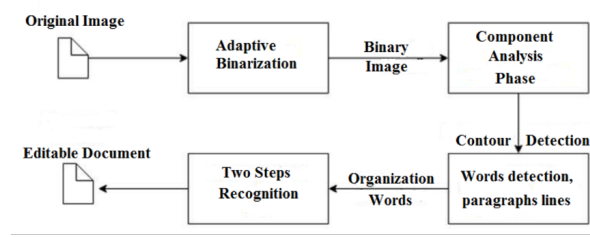


Figure 4: Tesseract OCR Architecture (Smith, R. 2007)

Nazeem et al. (2024) [7] studied OCR engines on low-resource languages. Though focused on Indian scripts, the results apply to Khmer. Their findings highlight Tesseract’s moderate accuracy and high adaptability when fine-tuned. A comparative performance chart from their study illustrates the performance gap and potential across scripts.

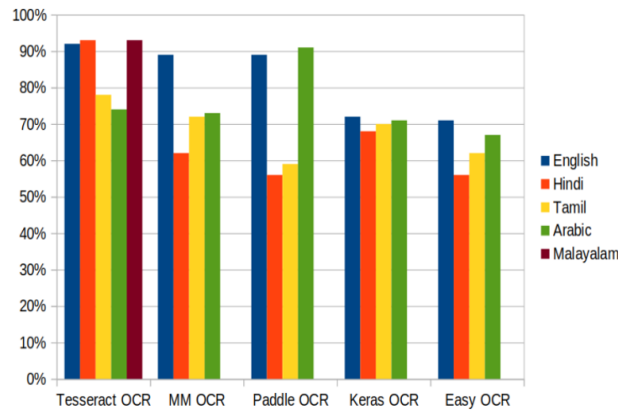


Figure 5: Comparison of OCR Library Performance in Different Languages [7]

2.4 Deep Learning Approaches to Khmer OCR

The rise of deep learning has significantly improved OCR systems for complex scripts like Khmer which traditional methods struggled to handle due to intricate structures and diacritics. Annanurov and Noor (2021) [1] addressed this by developing a compact OCR model for handwritten Khmer that combines CNN and RNN layers achieving 93–98% accuracy while remaining lightweight and suitable for low-resource environments.

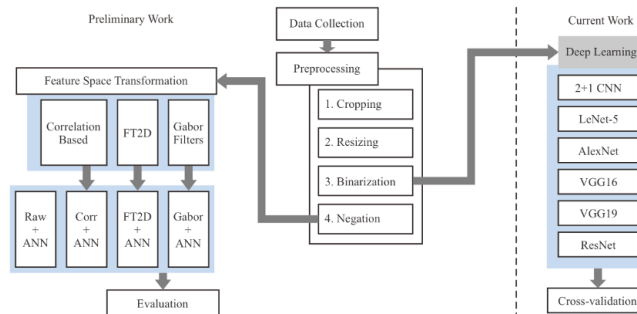


Figure 6: Research Framework [1]

Sin (2023) [11] analyzed Khmer invoice OCR using CNNs, CNN-RNN hybrids and Transformer-based models, finding that while CNNs effectively captured spatial features, Transformers outperformed in handling complex layouts and long-range dependencies. Modern models increasingly incorporate attention mechanisms to dynamically focus on relevant image regions, improving recognition of overlapping or subscript characters. Although attention-based frameworks like Transformers and encoder-decoder models show promise their effectiveness for Khmer is still constrained by the scarcity of large annotated datasets. Future research should explore semi-supervised learning, synthetic data generation and efficient attention-based models to enhance Khmer OCR performance, particularly in low-resource environments.

2.5 Compact and Efficient OCR Models for Printed Khmer

As the need to digitize Khmer printed materials grows in sectors like education and government there is increasing demand for OCR systems that are both accurate and computationally efficient especially on low-resource devices. Traditional OCR engines often struggle with Khmer script complexity under such constraints, prompting researchers to develop compact models using CRNNs and lightweight Transformers that maintain high accuracy while reducing memory and processing requirements. Schmidt and Kamlah (2022) [10] demonstrated that combining synthetic data with techniques like dynamic quantization and model pruning can produce fine-tuned models that outperform generic engines like Tesseract in both accuracy and efficiency making them suitable for constrained environments. Similarly, Patel (2021) [9] presented a customizable OCR framework using Tesseract, OpenCV, and Pytesseract for multilingual and flexible text detection; while not Khmer-specific, the approach provides useful insights for building modular, compact OCR pipelines adaptable to Khmer script.

3 Methodology

3.1 Overview of the Methodology

This study fine-tunes the Tesseract OCR engine using the Khmer OS Siemreap font, with a focus on adjusting the `max.iterations` parameter. The training iteration count was increased from the default 10,000 to 15,000 to improve recognition accuracy, especially for overlapping vowels and complex characters.

Raising `max.iterations` enhances training by:

1. **Increasing Accuracy:** More iterations allow deeper learning of subtle character variations in the dataset.
2. **Better Handling of Difficult Cases:** Extended training helps the model better recognize complex or challenging text patterns.

Performance is monitored through training loss and validation accuracy, ensuring the additional iterations lead to meaningful gains.

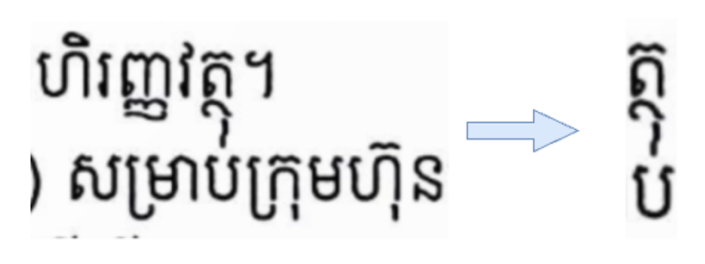


Figure 7: Overlapping Vowels with Diacritic

3.2 Dataset Comparison and Preprocessing

The dataset for this study was built by combining two Khmer text sources: the Khmer Text Recognition Dataset and `khmer-text-image-data`. It includes 9,315 high-resolution PNG images, totaling 136,117

words and 31,538 text lines. The goal was to focus on overlapping characters and diacritics—common issues in official documents using the Khmer OS Siemreap font.

Text content was first extracted into .txt files, then converted into .tif and .box files using the Khmer OS Siemreap font. These were used to generate .lstmf files for fine-tuning Tesseract’s LSTM-based OCR engine.

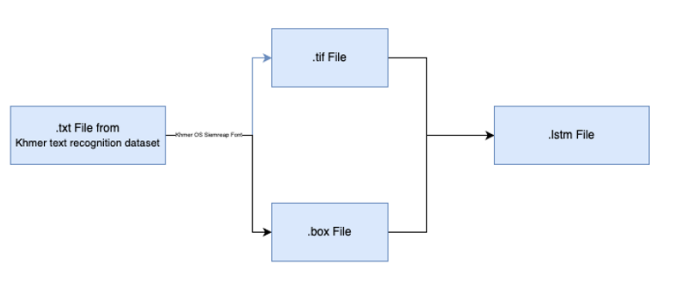


Figure 8: Flow of converting Training Dataset From (.txt) to (.lstm)

The use of a single, widely-used font allowed for controlled testing of OCR performance on difficult character arrangements such as stacked consonants and overlapping vowels. The dataset preparation followed these steps:

1. Extracted 136,117 words into .txt format from Khmer Text Recognition Dataset.
2. Collected 31,538 text lines from khmer-text-image-data.
3. Merged both sources into one .txt file.
4. Used the text2image tool to generate images with a white background and black Khmer text using the Siemreap font.

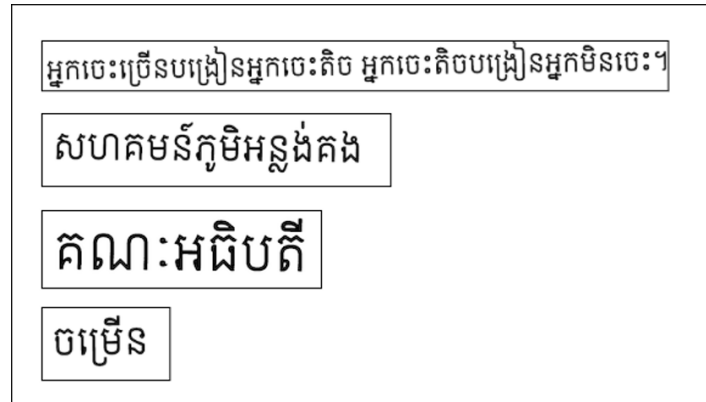


Figure 9: Sample Dataset use for Training

3.3 Max Iterations Method

Max Iterations in Tesseract LSTM training define the number of times for the model updates its weights to minimize the errors. A higher iterations count allows the model to refine the predictions to improving OCR accuracy especially for complex character structures like overlapping Khmer vowels.

The number of max_iterations is an important number for training. Moreover, max_iterations has their exact formula uses to calculate.

$$I = k \times D \times \frac{1}{LR}$$

Where:

I = max.iterations

D = number of training samples

LR = learning rate

k = a proportionality constant based on error convergence

To get more of understanding the formula of Max Iterations the number of max iterations (I), depends on dataset size (D), learning rate (LR) and model complexity.

Below of this we compare the different between max.iterations 10,000 with max.iterations 15,000. For Max Iterations = 10,000, D = 9,315 number of datasets for training and assuming a constant k and LR.

$$I_{10K} = k \times 9,315 \times \frac{1}{LR}$$

For Max Iterations = 15000, increasing iterations while keeping another factory constant.

$$I_{15K} = 1.5 \times I_{10K} = 1.5 \times k \times 9,315 \times \frac{1}{LR}$$

Increasing training iterations by 50% lets the model train longer for improving accuracy on overlapping vowels. More cycles mean the model learns the dataset better, leading to improved understanding of complex Khmer script and overlapping characters, vowels or diacritics.

3.4 Data Augmentation

Data augmentation improves OCR robustness by simulating real-world variations like misalignment, brightness changes, and geometric distortions. For Khmer script which has complex overlapping vowels and characters, augmentation helps the model focus on relevant text features while ignoring noise.

This study applies synthetic overlapping vowel augmentation by shifting vowels 20 positions to create overlaps, simulating spacing errors common in Khmer text. This trains the model to better recognize and separate overlapping vowels, enhancing accuracy and robustness in recognizing complex Khmer scripts, especially in scanned or difficult documents.

3.5 Fine-Tuning Tesseract LSTM Model

The study fine-tuned the Tesseract LSTM model manually without using Tesstrain for better control and customization tailored to Khmer script. Text files were converted to TIFF images with the Khmer OS Siemreap font using text2image. Corresponding BOX files with character-level annotations were generated using Tesseract's -psm 6. TIFF and BOX pairs were used to create .lstmf training files.

A starter model extracted from the original Khmer OCR served as the initial checkpoint. Training was done with increased iterations (15,000 vs. 10,000) to improve accuracy on complex vowels and overlapping glyphs. Key hyperparameters like learning rate and evaluation intervals were tuned, with periodic validation to track progress. The final model was recombined into a deployable .traineddata file. This manual pipeline enhanced OCR accuracy without relying on abstraction tools.

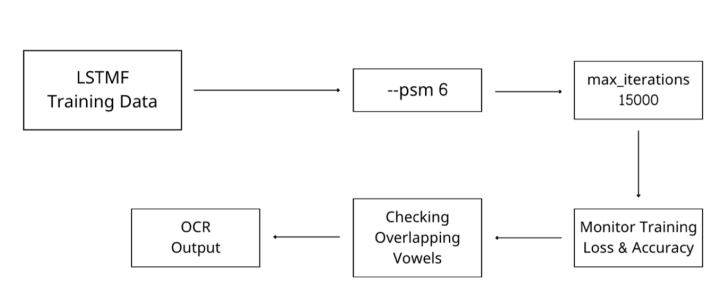


Figure 10: Training Process Workflow (Iterations & PSM Focus)

- The fine-tuning was performed using the Tesseract LSTM training framework on a macOS system with an Apple M1 chip.

3.7 Result Comparison and Discussion

Overlapping vowels in Khmer script present significant challenges for OCR due to their placement around consonants, often leading to segmentation and recognition errors (Muaz & LengIeng, 2015). These complexities can cause misalignment during character extraction, reducing accuracy. Buoy et al. (2021) [2] addressed this issue using a Sequence-to-Sequence model with attention, achieving a CER of 0.7% on a multi-font dataset. In contrast, this study’s fine-tuned Tesseract model, trained specifically on the Khmer OS Siemreap font, achieved a lower CER of 0.32% under controlled conditions, demonstrating the effectiveness of font specialization. Similarly, Sok and Taing (2014) [12] achieved 98.62% accuracy using SVMs on the Khmer OS Content font at 32pt size, while the fine-tuned Tesseract model in this study reached 99.68% accuracy at the same size, showing superior performance with overlapping characters. Further evaluation using 13,965 characters from 8 online newspaper articles resulted in a CER of 0.35% (99.65% accuracy), confirming the model’s robustness and real-world applicability. These results validate that fine-tuning Tesseract on a controlled, font-specific dataset can outperform more complex models in focused scenarios.

Type of Characters	Amount
Consonants	7995
Vowels	3535
Subscript	1136
Symbols	566
Diacritics	425
Number	224
Independent	55

Table 1: Dataset for Testing.

Increasing the number of max.iterations from 10,000 to 15,000 led to a significant improvement in OCR accuracy, particularly in recognizing overlapping vowels. The fine-tuned Tesseract model achieved a lower Character Error Rate (CER) showing enhanced recognition capabilities compared to the default Khmer Tesseract model with fewer errors in vowel positioning. These improvements were supported by comparative analysis tables illustrating reduced recognition errors. While increasing iterations clearly boosts accuracy the discussion also notes that further refinements in training data and preprocessing may be required for optimal performance. Additionally, the study acknowledges trade-offs such as longer training time and higher computational demands.

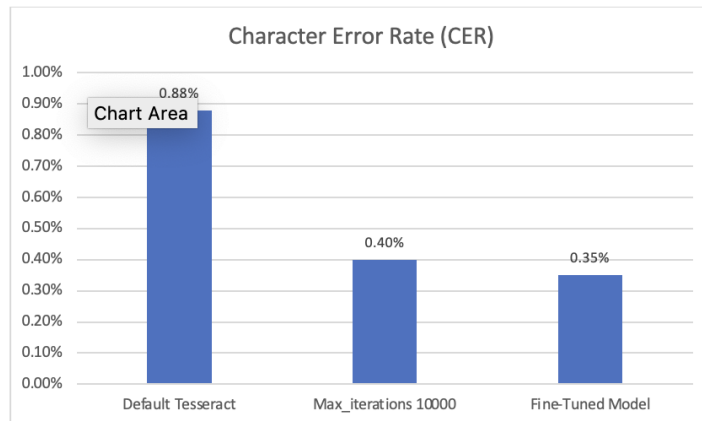


Figure 12: Result CER comparison with default tesseract, iterations 10000 and Fine-Tuned model

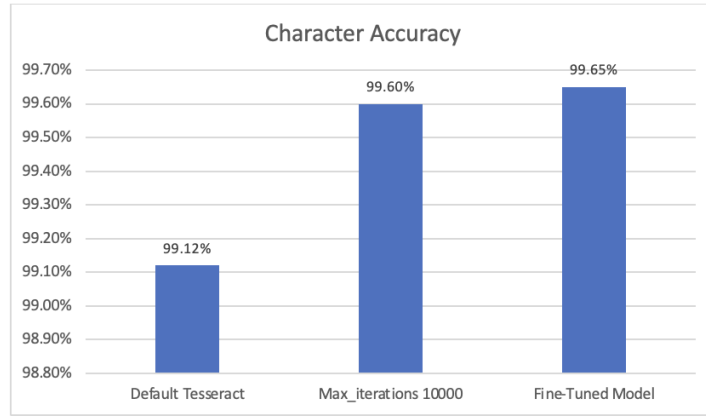


Figure 13: Result Character Accuracy comparison with default tesseraact, iterations 10000 and Fine-Tuned model



Figure 14: Testing overlap vowel with diacritic between default tesseraact, iterations 10000 and fine-tuned model

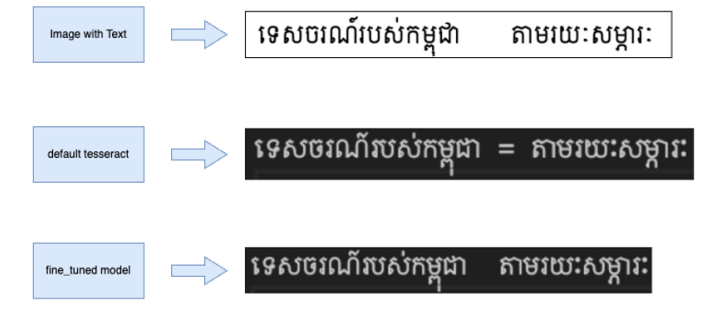


Figure 15: Testing dataset with long space

The histograms of Levenshtein distances on the validation set for the Fine-Tuned model, Iterations 10000 and Default Tesseract are given in figure 14 and figure15, respectively. A few observations can be made:

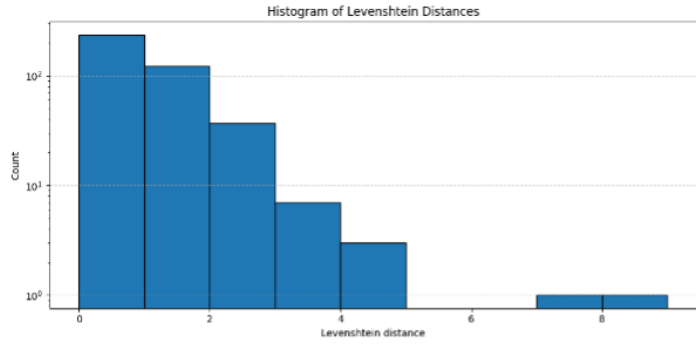


Figure 16: The Fine-tuned model histogram of Levenshtein distances on the validation

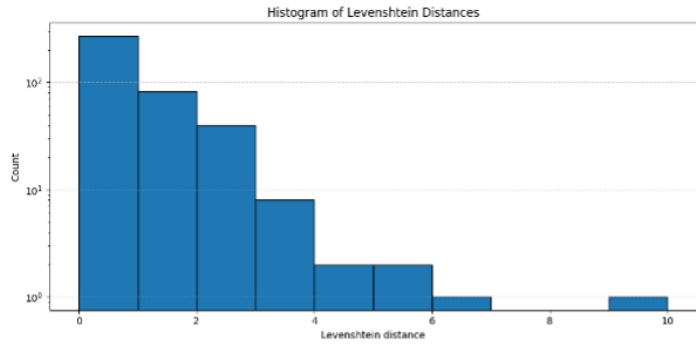


Figure 17: The Default Tesseract histogram of Levenshtein distances on the validation

One of the different between default tesseract, iterations 10000 and fine-tuned model is the accuracy or performance. Although, the time we spend on training more than iterations 10000 but the performance we achieved unexpected. One of an exact sample using max_iterations 15000 lead the model has more training cycles 50% but we can get the best performance on text which have overlapping vowels if we compare to default tesseract and iterations 10000.

3.8 Text Extraction from the Camera Image

The fine-tuned model not even can detect just only printed documents but for image of text which take from smartphone also can apply but in condition the image is clear and without noise background or without blur text in the image.

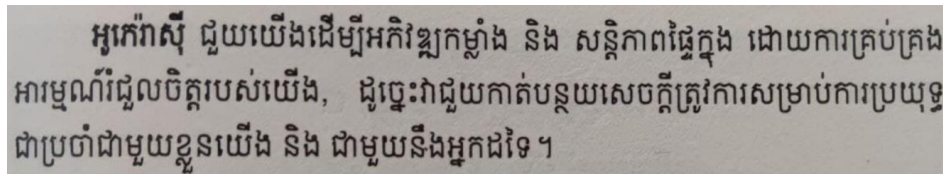


Figure 18: Another sample Khmer text from a smart phone's camera (Rina Buoy et al.2021) [2]

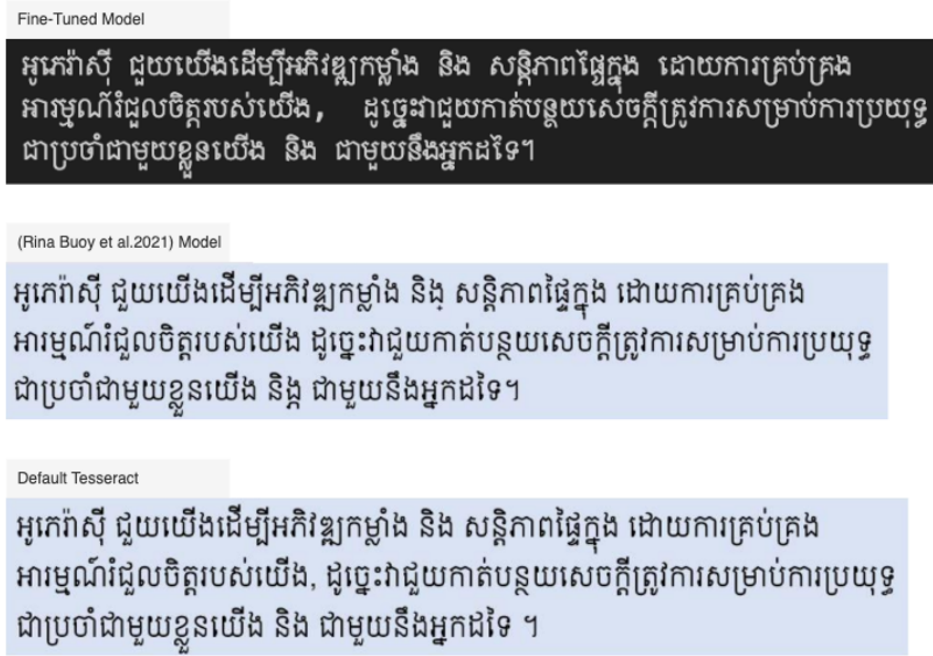


Figure 19: The recognized text by the Fine-Tuned model with (Rina Buoy et al.2021) [2] model and Default Tesseract

3.9 Overall Workflow

The overall workflow begins with data preparation, where Khmer text is converted into TIFF and BOX files using the Khmer OS Siemreap font, focusing on overlapping vowels and complex script structures. These are combined into LSTMF files for training the LSTM-based Tesseract OCR model. The fine-tuning process uses `-psm 6` and increases `max_iterations` from 10,000 to 15,000 to improve recognition accuracy. The model is then evaluated using Character Error Rate (CER) by comparing outputs to ground truth and the baseline model, with special attention to overlapping vowel cases. The final stage involves performance analysis and error categorization (substitution, insertion, deletion), demonstrating improved accuracy while identifying remaining challenges. This structured workflow provides a systematic approach to enhancing Khmer OCR performance.

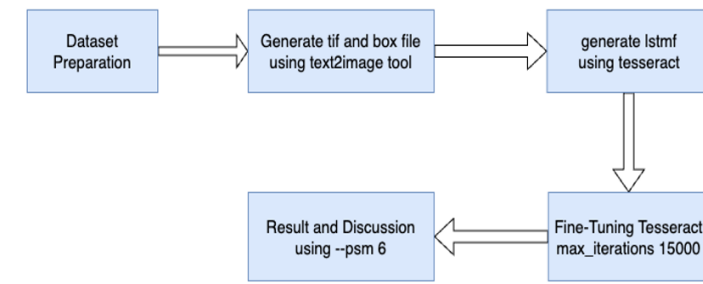


Figure 20: The overall workflow

4 Result and Discussion

4.1 Practical Implication

The fine-tuned Tesseract OCR model for the Khmer OS Siemreap font shows significant improvements in recognizing complex Khmer scripts, particularly overlapping vowels. These advancements are highly applicable to real-world scenarios such as digitizing historical and governmental documents in Cambodia, where many records contain challenging script structures. By improving recognition accuracy, the

enhanced model supports more effective digitization, preservation, and automation efforts, enabling institutions like national libraries and government agencies to convert printed archives into searchable, editable digital formats.

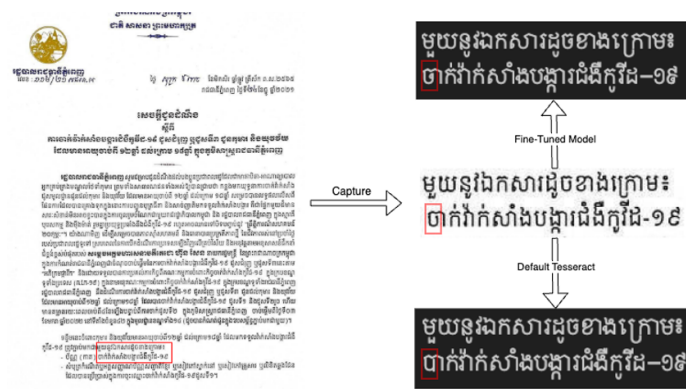


Figure 21: OCR Output Comparison on a Real Khmer Government Document Using Fine-Tuned Model vs Default Tesseract

4.2 Limitations of the Model

Although the fine-tuned Tesseract model significantly improved recognition of Khmer text, especially characters with overlapping vowels, several limitations remain. The model occasionally misinterprets spacing, merging or splitting words incorrectly—particularly when vowel diacritics extend into adjacent boundaries—due to the script’s dense vertical and horizontal character arrangements. Persistent errors also occur with certain vowel combinations involving subscript consonants, where minor glyph variations lead to misclassification. Additionally, while the 9,315-image dataset helped improve accuracy, it may not fully represent the range of Khmer script styles, print qualities, and distortions found in real-world documents. Prior research by Kruiy and Kameyama (2010) achieved an F-measure of 0.75 using SIFT features and a 96% hit rate with Connected Component (CC) recognition, suggesting room for further improvement. Increasing max_iterations from 10,000 to 15,000 lowered the Character Error Rate (CER), but the model still struggles with highly complex structures, indicating that more extensive training data and advanced segmentation techniques may be needed.

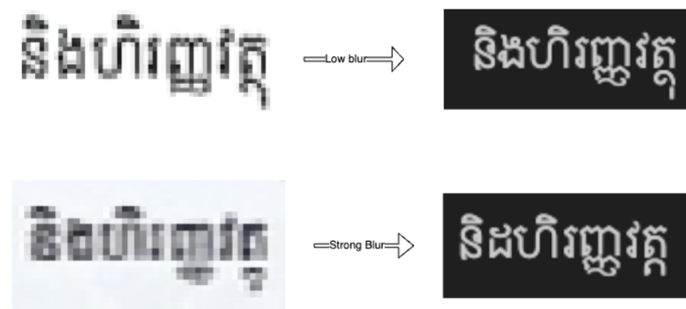


Figure 22: Compare result between blur image of text

4.3 Areas for Improvement

Despite the significant progress made by the fine-tuned Tesseract OCR model in recognizing Khmer script particularly overlapping vowels several limitations remain. The model was trained on a single font (Khmer OS Siemreap), which limits its ability to generalize across diverse fonts and real-world layouts such as multi-column formats, stylized headers, or low-resolution scans. Although synthetic vowel augmentation helped improve performance, additional augmentation techniques (e.g., rotation, blur, brightness shifts, and noise injection) could further enhance robustness. Hyperparameter tuning

was limited to increasing max_iterations; optimizing other parameters like batch size and learning rate could lead to further gains. Post-processing methods such as language models or dictionary-based corrections were also absent, which could otherwise improve error correction and spacing accuracy. Lastly, the model does not support multilingual or multi-font inputs, reducing its applicability in digitizing documents that mix Khmer and Latin scripts common in Cambodian content.

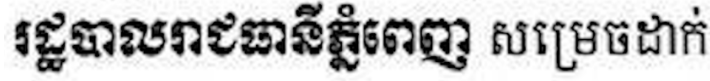


Figure 23: Sample khmer text with font variation



Figure 24: Fine-tuned results from khmer text with font variation

4.4 Future Work and Research Directions

This study shows that fine-tuning Tesseract for Khmer script especially overlapping vowels improves recognition accuracy, but further enhancements are needed for broader real-world applicability. Future work should include training on a variety of popular Khmer fonts (e.g., Khmer OS Battambang, Du-anPenh) and expanding the dataset to cover diverse text types like legal and educational documents. Introducing noise, brightness variation and distortion will help simulate real-world conditions and increase model robustness. Evaluating on scanned documents with imperfections such as blur and skew is also crucial. Additionally, exploring advanced OCR architectures like Convolutional Recurrent Neural Networks (CRNNs) which handle variable-length sequences without explicit character segmentation, could offer improved performance for complex Khmer scripts.

5 Conclusion

This study demonstrated that fine-tuning the Tesseract OCR engine using the Khmer OS Siemreap font and increasing training iterations from 10,000 to 15,000 significantly improved recognition accuracy achieving 99.65% character accuracy and a Character Error Rate (CER) of 0.35%. The model effectively addressed overlapping vowels, a common challenge in Khmer script. However, limitations remain in handling low-resolution images, diverse fonts and mixed scripts. Future work should expand font diversity, apply advanced OCR models (e.g., Transformers) and integrate language models. Building public Khmer OCR benchmarks and lightweight mobile-ready models will further enhance accessibility and real-world impact. This study lays a strong foundation for advancing Khmer OCR systems.

References

- [1] B. Annanurov and N. M. Noor. A Compact Deep Learning Model for Khmer Handwritten Text Recognition. *IAES International Journal of Artificial Intelligence*, 10(3):584–591, 2021.
- [2] R. Bouy, S. Kor, and N. Taing. An End-to-End Khmer OCR Using Sequence-to-Sequence with Attention., 2021. arXiv preprint arXiv:2106.10875.
- [3] R. Buoy and S. Chan. Khmer printed character recognition using an attention-based seq2seq network. In *Proceedings of the 2022 International Conference on Asian Language Processing (IALP)*, pages 1–6. IEEE, 2022.

- [4] R. Buoy, M. Iwamura, S. Srun, and K. Kise. Towards a Low-Resource Non-Latin Complete Baseline: An Exploration of Khmer OCR. In *Proceedings of IEEE BigComp 2023*, pages 199–206. IEEE, 2023.
- [5] V. Kruij and W. Kameyama. A Comparison with 1D Fourier Descriptor and Generic Fourier Descriptor on Khmer OCR. In *Proceedings of the 10th Information Science and Technology Forum (FIT 2011)*, 2011.
- [6] C. J. Mahoney, K. Jensen, H. Zhao, F. Wei, H. Qin, and S. Ye. Application of Deep Learning in Recognizing Bates Numbers and Confidentiality Stamping from Images. In *2020 IEEE International Conference on Big Data (Big Data)*, pages 4325–4332. IEEE, 2020.
- [7] M. Nazeem, R. Anitha, S. Navaneeth, and R. R. Rajeev. Open-Source OCR Libraries: A Comprehensive Study for Low-Resource Languages. In *Proceedings of the 21st International Conference on Natural Language Processing (ICON 2024)*, 2024.
- [8] V. Nom, S. Bakkali, M. M. Luqman, M. Coustaty, and J.-M. Ogier. Khmerst: A Low-Resource Khmer Scene Text Detection and Recognition Benchmark, 2024. arXiv preprint arXiv:2410.18277.
- [9] J. A. Patel. Handwritten and Printed Text Recognition Using Tesseract-OCR. *International Journal of Creative Research Thoughts (IJCRT)*, 9(3):2321–2326, 2021.
- [10] T. Schmidt and J. Kamlah. Finetune Your OCR! Improving Automated Text Recognition for Early Printed Works. Technical report, University Library Mannheim, 2022.
- [11] S Sin. Application of Deep Learning in Khmer Invoice Document Analysis. *Proceedings of the 3rd International Conference on Artificial Intelligence, Robotics, and Communication*, 2023.
- [12] P. Sok and N. Taing. Support Vector Machine (svm) Based Classifier for Khmer Printed Character-Set Recognition. In *2014 Asia-Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA)*, pages 1–9. IEEE, 2014.
- [13] K. Sokphyrum, S. Samak, and J. Sola. Khmer OCR Finetune Engine for Unicode and Legacy Fonts Using Tesseract 4.0 with Deep Neural Network. In *Optical Character Recognition for Complex Scripts and Natural Language Processing for ASEAN Languages*. 2019.
- [14] D. Valy, M. Verleysen, S. Chhun, and J.-C. Burie. A New Khmer Palm Leaf Manuscript Dataset for Document Analysis and Recognition: Sleukrith Set. In *Proceedings of the 4th International Workshop on Historical Document Imaging and Processing*, pages 1–6, 2017.